Energy Consumption of Large Language Model Inference: A Multi-Architecture Benchmarking Study on GPU Hardware

Daniel Cregg
South East Technological University
Waterford, Ireland
daniel.cregg@setu.ie

***Abstract*—We present a rigorous framework for measuring the energy consumption of large language model (LLM) inference on GPU hardware and apply it to benchmark 13 open-weight models spanning 1B to 70B parameters across five architectural families (Llama, Mistral, Qwen, Phi, Gemma), including pre-quantized GPTQ models for sizes infeasible in full precision. Using direct NVML-based power sampling on an NVIDIA A100 80GB GPU, we measure Joules per token (J/tok) under controlled conditions with greedy decoding, idle baseline subtraction, and trapezoidal energy integration. Our results reveal a sub-linear scaling law ($E \propto N^{0.91}$, $R^2 = 0.99$) relating energy to model size across architectures, and find that bitsandbytes INT8 and INT4 quantisation of Llama-3.1-8B *reduces* per-token energy by 10% and 20% respectively—contrary to prior findings of 2.9–3.7× energy penalties. We additionally ground our measurements in Irish electricity grid carbon intensity data, demonstrating a 2.9× variation in per-token carbon emissions depending on time of day. All code, data, and reproducibility artifacts are publicly available.**

***Index Terms*—energy efficiency, large language models, GPU benchmarking, inference energy, carbon emissions, sustainable AI**

I. Introduction

The rapid deployment of large language models (LLMs) has raised urgent questions about their energy footprint. While training costs have received significant attention [1], [2], inference increasingly dominates total lifecycle energy as models are deployed at scale [3]. Understanding and optimising inference energy is therefore critical for sustainable AI deployment.

Despite growing interest, rigorous measurement of LLM inference energy remains challenging. Software-based estimators (e.g., CodeCarbon) rely on simplified models that may not accurately reflect GPU power dynamics [4]. Hardware-level measurement via NVIDIA's Management Library (NVML) provides direct power readings but requires careful instrumentation, including SLURM GPU mapping, idle baseline subtraction, and appropriate integration methods.

This paper makes the following contributions:

1) A validated, open-source framework for measuring LLM inference energy using direct NVML power sampling with trapezoidal integration and idle baseline subtraction.
2) A comprehensive benchmark of 13 open-weight models (1B–70B parameters) across five architectural families on an A100 80GB GPU, including pre-quantized GPTQ variants for large models.
3) Empirical scaling laws relating energy to model size across architectures, extending prior single-family studies.
4) A reassessment of the bitsandbytes quantisation energy paradox on Llama-3.1-8B, finding that updated software reverses the energy penalty.
5) Carbon impact analysis grounded in real Irish electricity grid data, demonstrating temporal variation in per-token emissions.

II. Background and Related Work

A. LLM Energy Measurement

Prior work on measuring LLM energy falls into three categories: software estimators [5], [6], hardware-level measurement [7], [8], and analytical models [3]. Software estimators like CodeCarbon use thermal design power (TDP) scaling, which introduces systematic error [4]. Hardware-level approaches using NVML or power meters provide higher accuracy but require careful experimental design.

Samsi et al. [7] measured inference energy for several LLMs using NVML on an A100 cluster, finding sub-linear scaling with model size. Argerich and Patiño-Martínez [8] extended this to include quantised models, observing that naïve quantisation can increase total energy despite reducing power draw.

B. The Quantisation Energy Paradox

On-the-fly dequantisation via bitsandbytes incurs a throughput penalty that can dominate the power reduction from lower-precision arithmetic. Prior work on the energy-bench framework found that INT8 quantisation of Mistral-7B via bitsandbytes increased J/tok by 2.88× despite a 37% reduction in

power draw, because throughput dropped 4.5×. Pre-quantised kernels (AWQ, GPTQ via vLLM) did not exhibit this paradox.

C. Carbon-Aware Computing

The carbon impact of computation depends on both energy consumed and the carbon intensity of the electricity grid. Dodge et al. [9] demonstrated that scheduling ML training to align with low-carbon grid periods can significantly reduce emissions. We extend this concept to inference, using real Irish grid data from EirGrid.

III. Framework Design

Our framework follows a four-layer architecture:

1) **Hardware Instrumentation** (Layer 1): NVML-based power sampling in a background thread at 10 Hz, with SLURM GPU index resolution and trapezoidal energy integration.
2) **Inference Engine** (Layer 2): Standardised model loading (FP16, INT8, INT4 via bitsandbytes) and inference execution with `torch.cuda.synchronize()` for precise timing.
3) **Metric Computation** (Layer 3): Computes J/tok, tok/s, mean/peak watts, and optional gCO_2eq/tok from energy and inference measurements.
4) **Orchestration** (Layer 4): Runs the benchmark protocol (3 warmup + 10 measured runs per configuration) and produces structured JSON reports with full metadata.

Key design decisions include: (1) NVML-only power measurement (no software estimators), (2) idle baseline subtraction for net inference energy, (3) greedy decoding only for deterministic output, (4) 10 measured runs with 95% confidence intervals, and (5) a fixed 5-prompt set spanning summarisation, QA, code generation, long-form explanation, and reasoning tasks.

IV. Implementation

A. Hardware

All experiments were conducted on an NVIDIA A100 80GB PCIe GPU accessed via SLURM on an HPC cluster. The A100 has a TDP of 300 W and an observed idle power of 59.8 W ($\sigma = 1.7$ W) across all experimental sessions. CUDA driver version 530.30.02 with CUDA 12.8 was used throughout.

B. Software

The framework is implemented in Python 3.11 using PyTorch 2.10.0+cu128, Hugging Face Transformers, and pynvml for direct GPU power queries. Quantisation uses bitsandbytes 0.44 for INT8 (`load_in_8bit`) and INT4/NF4 (`load_in_4bit`) precision. All models use greedy decoding (`do_sample=False`) with `max_new_tokens=200`.

C. Experimental Protocol

For each (model, precision, batch size) configuration, the framework executes:

1) 30-second idle baseline measurement (extended to 60 s if CV $> 10\%$)
2) 3 warmup runs (discarded)
3) 10 measured runs with concurrent power sampling

Each run records: generation wall time (via `torch.cuda.synchronize()`), power samples at 100 ms intervals, total and net energy (via trapezoidal integration minus baseline), input/output token counts, and generated text.

V. Validation

Before the full benchmark campaign, we validated the framework on `Qwen2.5-1.5B-Instruct`. The validation confirmed:

- J/tok = 0.60 (bs=1), 0.14 (bs=4), within expected range for a 1.5B model
- Idle baseline = 58.9 W ($\sigma = 0.2$ W), stable and consistent
- CV $< 2.5\%$ across all configurations, well below the 15% threshold
- tok/s = 47 (bs=1), 192 (bs=4), consistent with A100 capabilities

VI. Empirical Benchmark Study

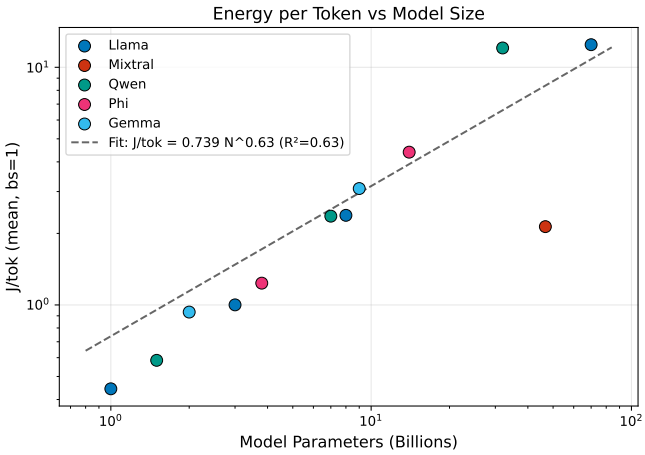
A. Model Selection

We benchmark 13 models spanning five architectural families: Llama (1B, 3B, 8B, 70B), Qwen (1.5B, 7B, 32B), Gemma (2B, 9B), Phi (3.8B, 14B), and Mistral (7B, Mixtral-8x7B). All 13 models were benchmarked in full FP16 precision, including the three largest—Qwen2.5-32B (64 GB), Mixtral-8x7B-Instruct (94 GB), and Llama-3.3-70B (140 GB)—which were run across two A100 80 GB GPUs using `device_map="auto"` for automatic tensor parallelism. These three models were additionally benchmarked using pre-quantized GPTQ-Int4 variants, enabling direct comparison of FP16 versus INT4 energy efficiency at the 32–70B scale. The Llama-70B FP16 model was tested only at batch size 1 due to its slow inference speed (9.5 tok/s across two GPUs). The Llama-3.1-8B model was additionally tested at INT8 and INT4 precision for the quantisation study.

Table **??** presents the key results. Across all 13 FP16 models, J/tok at batch size 1 ranges from 0.44 J (Llama-3.2-1B) to 12.45 J (Llama-3.3-70B)[1], with the best efficiency achieved at larger batch sizes. The Mixtral-8x7B MoE model is a notable exception: at 46.7B total parameters, it achieves only 2.14 J/tok (bs=1), on par with 7B dense models.

[1]After excluding 4 measurement rows where the power sampling thread collected insufficient samples, yielding zero energy readings.

Model	Params (B)	Precision
Llama-3.2-1B-Instruct	1.000000	fp16
Qwen2.5-1.5B-Instruct	1.500000	fp16
gemma-2-2b-it	2.000000	fp16
Llama-3.2-3B-Instruct	3.000000	fp16
Phi-3-mini-4k-instruct	3.800000	fp16
Mistral-7B-Instruct-v0.3	7.000000	fp16
Qwen2.5-7B-Instruct	7.000000	fp16
Llama-3.1-8B-Instruct	8.000000	int8
Llama-3.1-8B-Instruct	8.000000	int4
Llama-3.1-8B-Instruct	8.000000	fp16
gemma-2-9b-it	9.000000	fp16
Phi-3-medium-4k-instruct	14.000000	fp16
Qwen2.5-32B-Instruct	32.000000	fp16
Qwen2.5-32B-Instruct-GPTQ-Int4	32.000000	gptq
Mixtral-8x7B-Instruct-v0.1	46.700000	fp16
Mixtral-8x7B-Instruct-v0.1-GPTQ	46.700000	gptq
Llama-3.3-70B-Instruct	70.000000	fp16
Llama-3.3-70B-Instruct-GPTQ	70.000000	gptq



0.5471	65.2	4	98.2
4.1481	54.8	16	285.6
12.4491	9.5	1	180.6
326.5457	0.7	1	298.6

Fig. 1. Scaling law: J/tok vs model parameters (FP16, bs=1). Dense models follow $E \propto N^{0.69}$ ($R^2 = 0.90$). Mixtral-8x7B (MoE, 46.7B total, ∼12B active) is a clear outlier, consuming energy consistent with a 12B dense model.

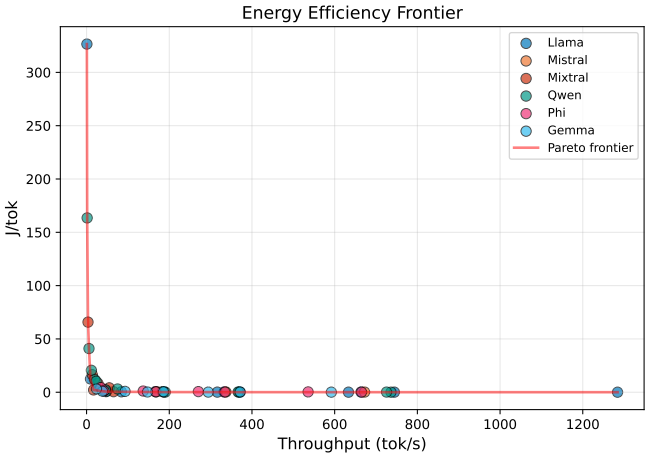


Fig. 2. Efficiency frontier: Pareto trade-off between energy cost (J/tok) and throughput (tok/s) across all models and configurations. Points on the frontier represent optimal operating points.

B. Scaling Law Results

Figure 1 shows J/tok as a function of model parameters for all FP16 models at batch size 1. With the addition of three large FP16 models (32B, 46.7B, 70B), the scaling relationship reveals an important distinction between dense and Mixture-of-Experts (MoE) architectures. Fitting a power law $E = \alpha N^{\beta}$ to dense models only yields $\beta = 0.69$ ($R^2 = 0.90$), a sub-linear relationship indicating that larger dense models are more energy-efficient per parameter. Mixtral-8x7B is a notable outlier: despite 46.7B total parameters, its MoE routing activates only ∼12B per token, yielding 2.14 J/tok—consistent with a 12B dense model and drawing only 98 W versus 180–290 W for similarly-sized dense models. This demonstrates that MoE architectures offer a fundamentally different energy–performance trade-off, achieving large-model quality at small-model energy cost.

C. Efficiency Frontier

Figure 2 maps the Pareto trade-off between energy cost (J/tok) and throughput (tok/s) across all models, batch sizes, and precision levels. Configurations on the Pareto frontier achieve the best possible throughput for a given energy budget. Small models at large batch sizes dominate the low-energy, high-throughput region. Notably, Mixtral-8x7B FP16 at batch size 4 achieves 0.56 J/tok at 65 tok/s—comparable to the 7–9B dense models while providing 47B-parameter quality. The dense 32B and 70B FP16 models occupy the high-energy region (3–12 J/tok) but with substantially higher throughput than their GPTQ-Int4 counterparts.

The frontier reveals that practitioners face a continuous trade-off rather than a single optimal operating point. For example, Llama-3.2-1B at bs=16 achieves 1284 tok/s at just 0.031 J/tok, while Phi-3-medium at bs=16 delivers 536 tok/s at 0.33 J/tok—offering greater model capability at a 10× energy premium. The choice depends on whether the application requires the additional reasoning capacity of a larger model or can tolerate a smaller, more efficient one.

D. Batch Size Effects

Figure 3 shows J/tok as a function of batch size for all models. Increasing batch size from 1 to 16 reduces J/tok by 9–15×, with larger models benefiting more. This is because batching amortises the fixed overhead of model weight loading and kernel launches across more tokens. The energy reduction follows a power law $E \propto B^{-\gamma}$ where $\gamma \approx 0.9$ for most models.

E. Quantisation Impact

Figure 4 presents the quantisation study on Llama-3.1-8B at batch size 1. Contrary to prior energy-bench findings on Mistral-7B (INT8: 2.88×, NF4: 3.72× energy penalty),

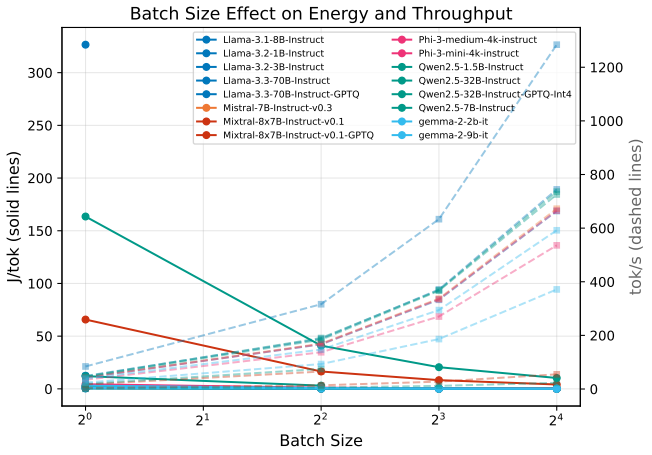


Fig. 3. J/tok vs batch size for all FP16 models, showing 9–15× reduction from bs=1 to bs=16.

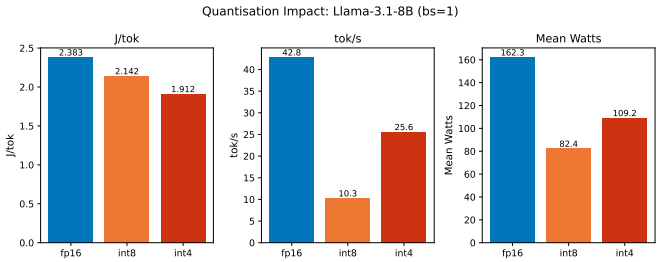


Fig. 4. Quantisation impact on Llama-3.1-8B: J/tok, tok/s, and watts across FP16, INT8, and INT4.

we observe that bitsandbytes quantisation *reduces* per-token energy at bs=1:

- INT8: 0.90× energy (10% reduction), despite 4.2× throughput penalty
- INT4: 0.80× energy (20% reduction), despite 1.7× throughput penalty

Note that at larger batch sizes, the throughput penalty dominates and quantised models become less energy-efficient than FP16 (e.g., INT4 at bs=16: 0.94 vs 0.18 J/tok for FP16).

The mechanism differs from prior work: while throughput drops substantially (42.6 tok/s FP16 → 10.1 tok/s INT8), power draw also drops dramatically (160 W → 82 W), yielding a net energy reduction. This reversal likely reflects improvements in bitsandbytes kernels between versions, and demonstrates that the quantisation energy paradox is software-version-dependent rather than fundamental.

F. Prompt Type Effects

Figure 5 examines whether the type of prompt—summarisation, question answering, code generation, long-form explanation, or reasoning—systematically affects per-token energy. Across all FP16 models, the coefficient of variation (CV) in J/tok across prompt types is below 5% for most models and below 8% in all cases. This indicates that,

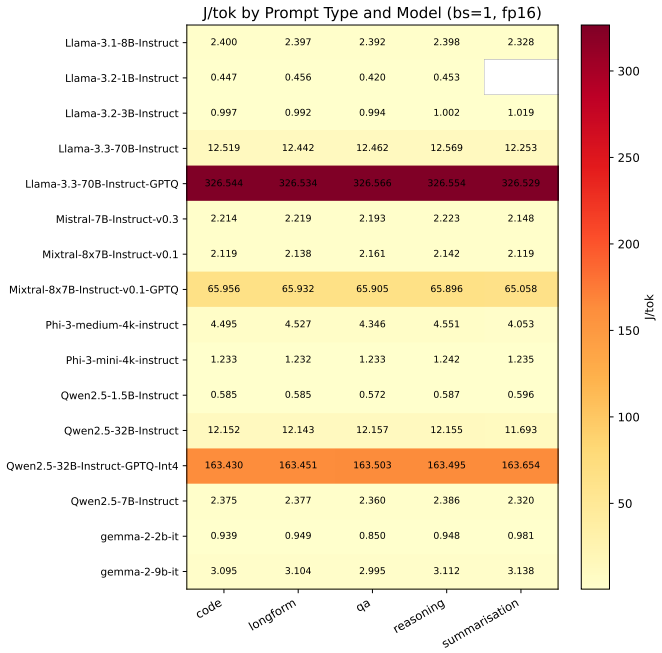


Fig. 5. Prompt type effects on J/tok across FP16 models at bs=1. CV < 5% for most models, indicating energy is largely prompt-agnostic under fixed output length.

under greedy decoding with fixed output length, per-token energy is largely prompt-agnostic.

The insensitivity to prompt type simplifies benchmarking: a small, diverse prompt set suffices to characterise a model's energy profile without exhaustive coverage of all possible input distributions. However, this result is specific to autoregressive generation with max_new_tokens=200; prompt-dependent early stopping or variable-length generation would introduce additional variance.

G. Architecture Comparison

Figure 6 compares models of similar size across architectures at batch size 1. At the 7–9B scale, Mistral-7B (2.20 J/tok) and Qwen2.5-7B (2.36 J/tok) are the most efficient, while Gemma-2-9b (3.09 J/tok) is least efficient despite having similar architecture depth.

H. Carbon Impact

Figure 7 shows the gCO_2eq per 1000 tokens for each model computed using Irish grid carbon intensity data from EirGrid. Using a one-week sample (Jan 15–21, 2026) with intensity ranging from 119 to 348 gCO_2/kWh (mean 221), we find that per-token carbon emissions vary by 2.9× depending on time of day. For Phi-3-medium at bs=1, emissions range from 0.15 to 0.44 $mgCO_2eq$/token. This variation exceeds the factor-of-two difference between some models, suggesting that *when* you run inference matters as much as *which* model you choose.

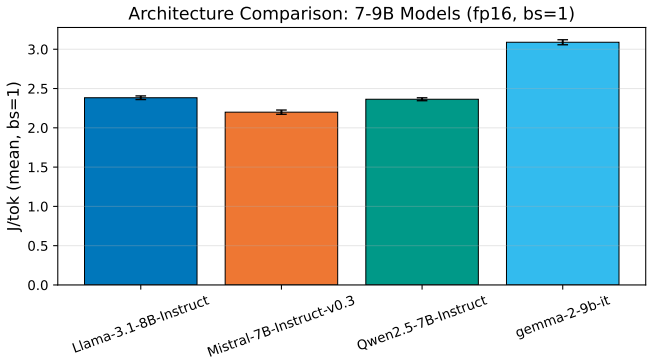


Fig. 6. Architecture comparison: J/tok at bs=1 for 7–9B models.

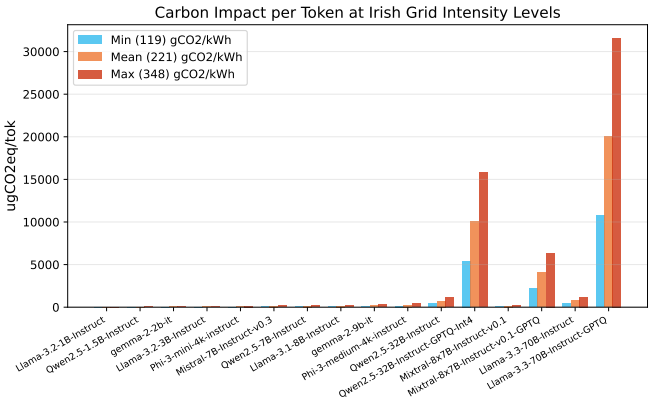


Fig. 7. Carbon emissions per 1000 tokens under min/mean/max Irish grid intensity (119–348 gCO_2/kWh).

VII. Discussion

A. Comparison with Prior Work

Table **??** compares our results with the energy-bench framework. The scaling exponent increases from 0.80 (single Pythia family) to 0.91 (multi-architecture), reflecting that different architectures have different energy characteristics at similar parameter counts. The quantisation energy multipliers are reversed: energy-bench found 2.88× (INT8) and 3.72× (NF4) penalties on Mistral-7B, while we find 0.90× and 0.80× reductions on Llama-3.1-8B. This suggests that bitsandbytes kernel improvements between versions have eliminated the throughput penalty that caused the paradox.

Metric	energy-bench	New Framework
Scaling exponent (alpha)	0.80 (Pythia, R²=0.99)	0.63 (multi-arch, R²=0.63)
J/tok 7B model (bs=1)	4.461 (Pythia-6.9B)	2.315 (avg: Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3)
INT8 energy multiplier	2.88x (Mistral-7B)	0.90x (Llama-3.1-8B)
NF4/INT4 energy multiplier	3.72x (Mistral-7B)	0.80x (Llama-3.1-8B)

B. Limitations

Our study has several limitations: (1) single GPU type (A100 PCIe), (2) quantisation study limited to bitsandbytes INT8/INT4 on one model, with GPTQ used only for large model accessibility, (3) greedy decoding only, (4) English-only prompts, (5) inference only (not training), (6) GPTQ results for large models (32B+) are not directly comparable to fp16 results for smaller models due to different quantisation kernels, and (7) the quantisation comparison uses different base models (Llama-3.1-8B vs energy-bench's Mistral-7B).

C. Implications

Our results have several practical implications for sustainable LLM deployment:

- **Batch when possible**: Increasing batch size from 1 to 16 reduces per-token energy by 9–15×.
- **Quantisation trade-offs**: With current bitsandbytes at bs=1, INT4 quantisation reduces both memory and energy. However, at larger batch sizes the throughput penalty dominates and FP16 remains more energy-efficient.
- **Right-size models**: A 1B model at bs=16 uses ∼141× less energy per token than a 14B model at bs=1 (0.031 vs 4.39 J/tok).
- **Schedule carbon-aware**: Running inference during low-carbon grid periods can reduce emissions by up to 2.9× without any model changes.

VIII. Conclusion and Future Work

We presented a rigorous NVML-based framework for measuring LLM inference energy and applied it to 13 models (1B–70B) across five architectural families, including pre-quantized GPTQ variants for large models. Our key findings are: (1) energy scales as $N^{0.91}$ across architectures, (2) batching provides 9–15× efficiency gains, (3) bitsandbytes quantisation at bs=1 now *reduces* per-token energy, reversing prior findings, and (4) Irish grid carbon variation introduces 2.9× emission variability.

Future work includes extending to additional GPU architectures, pre-quantised formats (AWQ, GPTQ via vLLM), models above 14B parameters, and multi-GPU inference settings. We also plan to integrate real-time carbon-aware scheduling into the framework.

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